

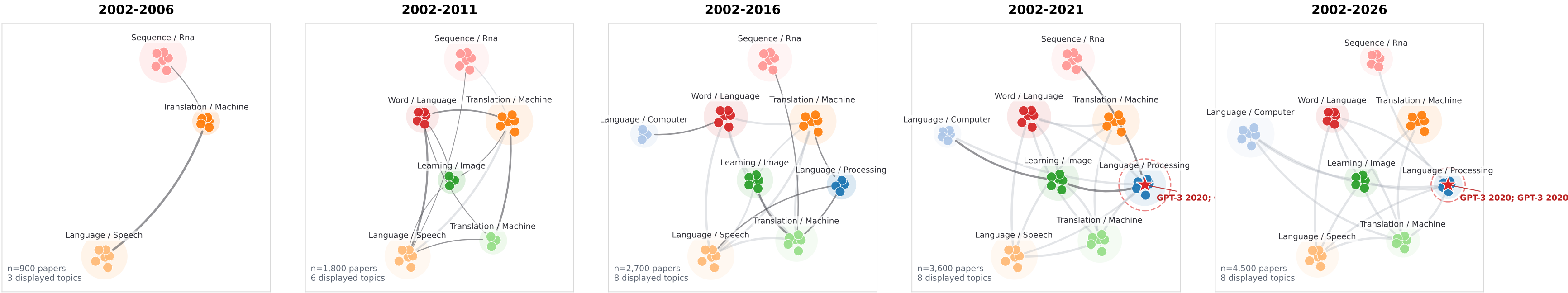
Fig. 1 | Citation knowledge-graph perturbation after a landmark innovation

Example domain: Transformer and foundation models. Real OpenAlex data; publication-style display uses data-derived topic snapshots.

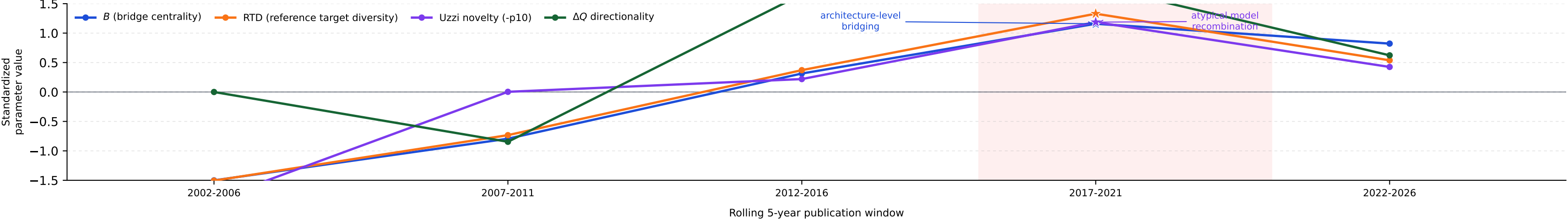


a Cumulative citation knowledge graph snapshots

Transformer and foundation models



b Dominant parameter trajectories explaining the graph transitions in panel a



Bridging
B up + RTD up
Architecture-level communities connect

Novelty
Uzzi up
Attention-only recombination is atypical

Standardization
Uzzi down; B high
Convergence into foundation-model hubs